Logistic Regression (Binary Classification)

i) Start

ii) Input: Training dataset with features X.
Target variable y (values 0 or 1)
Learning rate α.
No. of iterations T.

iii) Initialize weight vector w and bias b to zero.

iv) Repeat for $t = 1$ to T:

a) Compute linear combination : $z = wX + b$

b) Apply sigmoid function to obtain predicted probability

$$\hat{y} = \frac{1}{1+e^{-z}}$$

c) Compute error : $\text{error} = \hat{y} - y$

d) Compute gradient wrt weights: $\frac{\partial J}{\partial w} \leftarrow \frac{1}{n} X^T (\text{error})$

e) Compute gradient wrt bias: $\frac{\partial J}{\partial b} \leftarrow \frac{1}{n} \Sigma (\text{error})$

f) Update weights: $w \leftarrow w - \alpha \frac{\partial J}{\partial w}$

g) Update bias: $b \leftarrow b - \alpha \frac{\partial J}{\partial b}$

v) Classify output: if $\hat{y} \geq 0.5$, predict class 1.
otherwise, predict class 0.

vi) Output trained model parameters.

vii) End.